AI FDE Capstone — Engineering, Transformation & Final Presentation Guidelines

A. Capstone Engineering & Transformation Guidelines

#	Guideline	Expectations
1	42-Stage AI FDE Spine	Treat the 42-stage spine as an indicative yet prescriptive reference framework. Preserve its core disciplines, but adapt the sequence and depth to the use case. It is not written in stone.
2	Prompt Adaptability	Add, refine, combine, or modify prompts and steps based on domain complexity, discoveries, technical constraints, and business requirements.
3	Enterprise Context Lake	Build a structured, versioned, evidence-backed Context Lake capturing business rules, domain knowledge, architecture, code intelligence, data relationships, dependencies, and decisions.
4	Two Functional Applications	Build two independently functional applications—n AI-transformed application & Context Lake ported application
5	Enterprise-Grade UI/UX	Deliver professional, fully functional applications with realistic workflows, dashboards, navigation, integrations, role-based experiences, and exception handling. No half-baked prototypes or static mockups.

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6	Agentic AI Engineering	Implement meaningful agentic workflows with planning, tool orchestration, execution, context management, bounded autonomy, and human approval controls.
7	Knowledge Graphs	Explore and implement knowledge graphs for entity relationships, domain dependencies, contextual reasoning, and explainability wherever justified by the use case.
8	Semantic Layer	Establish a governed semantic layer with canonical business entities, standardized terminology, relationships, metrics, and cross-system mappings.
9	Specification-Driven Development (SDD)	Drive implementation through explicit requirements, specifications, API/data contracts, architectural decisions, constraints, and acceptance criteria.
10	Test-Driven Development (TDD)	Apply TDD with characterization, unit, integration, regression, contract, and end-to-end tests to ensure correctness and safe modernization.
11	Security & Threat Modeling	Conduct threat modeling across application, infrastructure, data, APIs, LLMs, RAG, agents, tools, and integrations. Implement and validate appropriate controls.
12	Compliance & Governance	Address applicable domain regulations, privacy, responsible AI, auditability, data governance, model governance, and enterprise compliance obligations.
13	Production Engineering	Ensure the following are considered - observability, resilience, failure recovery,

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		rollback, performance, scalability, and operational readiness.
14	Evidence-Based Comparison	Compare both applications using identical business scenarios and measurable KPIs covering accuracy, latency, productivity, reliability, cost, and ROI.
15	Birlasoft Branding	Use approved Birlasoft logos, corporate slide templates, brand guidelines, typography, and consistent professional styling.
16	Board-Level Presentation	Deliver an executive-ready presentation focused on business transformation, architectural choices, technical differentiation, risk mitigation, and measurable enterprise value.
17	Live End-to-End Demonstration	Demonstrate both working applications, including real business workflows, AI intelligence, agentic execution, exception handling, and human approvals—not screenshots alone.
18	Engineering Evidence & Traceability	Support critical architectural decisions, implemented capabilities, security controls, and business improvements with verifiable artifacts, tests, and execution evidence.

B. Final Presentation & Executive Defence Guidelines

#	Guideline	Expectations
19	Time-Boxed Presentation	Strictly 40 minutes for explanation and live demonstration + 10 minutes for Q&A. Maintain a focused agenda and eliminate unnecessary detail.

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20	Compelling Storytelling	Build a gripping, data-driven narrative connecting the business problem, its consequences, the transformation journey, and measurable outcomes.
21	Problem → Solution → Impact	Clearly articulate the problem, what is being solved, how the solution works, and the business value delivered.
22	Business + Technical Balance	Give equal importance to business context, measurable value, technical architecture, engineering decisions, and actual implementation.
23	Before vs. After KPIs	Present evidenced baseline and post-transformation KPIs covering business outcomes, engineering productivity, quality, performance, and reliability.
24	GenAI Value Justification	Explain why GenAI was necessary, which problems genuinely required it, and why conventional approaches alone were insufficient. Separate GenAI contributions from traditional engineering improvements.
25	Visual-First Communication	Minimize text. Prioritize architecture diagrams, business workflows, data visualizations, KPI comparisons, and live system demonstrations.
26	Evidence-Based Demonstration	Support major claims with proof of work: code changes, test results, execution traces, evaluation reports, benchmark outputs, and working functionality.
27	End-to-End Transformation Traceability	Demonstrate the journey from legacy discovery and business-rule extraction through architecture, modernization, AI

#	Guideline	Expectations
		integration, validation, and production readiness.
28	Crisp Lessons Learned	Conclude with 4–5 meaningful lessons covering engineering insights, business implications, GenAI limitations, unexpected challenges, and future improvements.

Mandatory Demo Quality Principles

- Show, don't just tell: Prioritize working demonstrations over slide explanations.
- Prove, don't just claim: Every improvement must have credible supporting evidence.
- Quantify, don't generalize: Use measurable KPIs with consistent definitions, comparable test conditions, and visible baseline results.
- Justify, don't force-fit GenAI: Clearly separate where GenAI adds differentiated value from where conventional software engineering is more appropriate.
- Connect, don't fragment: Every technical capability demonstrated should map to a business problem and measurable outcome.

By the end of the presentation, the audience should clearly understand the original business problem, the complexity of the legacy system, why and where GenAI was applied, how the transformation was engineered, and what measurable business and technical improvements were actually achieved.